

Phase 4 - DFN simulation

PyBaMM BasicDFNComposite with the published Chen2020_composite parameter set (LGM50 NMC811 / graphite-SiOx negative). Image inputs: measured porosity AND a literature-range correction (set to 0.30, literature range for calendered anodes; small pores unresolved), active fractions, particle radii. Under measured porosity the model predicts Batch_1 and Batch_2 lose ~3% capacity vs Batch_3 at 2C only (CI excludes zero); under the corrected-0.30 porosity the batches are indistinguishable - true porosity dominates the small image differences. Absolute values are NOT validated; only between-batch differences matter. The model has no swelling, cracking or ageing.

Assumption (applies throughout): the grey bulk is assumed graphite, bright particles are assumed silicon-based, black is pore/crack. This comes from image appearance and is unconfirmed.

image_id	batch	porosity	gr_amf	si_amf	gr_radius_m	si_radius_m
4ih2ggld	Batch_1	0.1134	0.718	0.1686	1.442e-06	2.631e-07
5n1q8atc	Batch_1	0.08157	0.7344	0.184	1.622e-06	2.593e-07
f1vzngrs	Batch_1	0.08005	0.8373	0.08262	2.801e-06	3.317e-07
ffwubibz	Batch_1	0.09383	0.8575	0.04862	3.562e-06	4.311e-07
fzrt2k6r	Batch_1	0.1061	0.8256	0.06837	2.935e-06	3.675e-07
iv6g2oq0	Batch_1	0.1116	0.8274	0.06101	3.189e-06	3.182e-07
uhdslk0o	Batch_1	0.1044	0.8305	0.0651	2.989e-06	3.119e-07
3806gxp0	Batch_2	0.1208	0.8187	0.06052	2.731e-06	3.317e-07
avn74qx1	Batch_2	0.1127	0.8427	0.04463	3.389e-06	4.343e-07
b3esycq1	Batch_2	0.1079	0.8044	0.08769	2.705e-06	3.469e-07
epqdaau9	Batch_2	0.09437	0.8381	0.06753	3.048e-06	3.036e-07
i9jiqjwl	Batch_2	0.1012	0.8247	0.07411	2.808e-06	3.33e-07
r17byphk	Batch_2	0.1246	0.8218	0.05361	2.743e-06	2.948e-07
rxax5ozo	Batch_2	0.0996	0.8425	0.05794	3.179e-06	3.174e-07
0grcilhi	Batch_3	0.1511	0.7836	0.06531	2.466e-06	3.042e-07
71vgq3fw	Batch_3	0.0944	0.833	0.07265	2.734e-06	3.079e-07
9luzk4jm	Batch_3	0.1499	0.789	0.0611	2.443e-06	4.87e-07
cfe5vt7s	Batch_3	0.1078	0.8218	0.07043	2.813e-06	3.284e-07
hawkfj64	Batch_3	0.1046	0.8463	0.04905	3.135e-06	3.426e-07
hzumfsms	Batch_3	0.1638	0.7861	0.05008	2.661e-06	4.678e-07

batch	porosity_mode	c_rate	n	cap_mean	cap_std	cap_p5	cap_p95
Batch_1	measured	0.1	30	5.25	8.456e-15	5.25	5.25
Batch_1	measured	0.5	30	5.25	5.927e-15	5.25	5.25
Batch_1	measured	1	30	5.25	3.626e-15	5.25	5.25
Batch_1	measured	2	30	3.813	0.1453	3.564	4.021
Batch_1	corrected	0.1	30	5.25	6.417e-15	5.25	5.25
Batch_1	corrected	0.5	30	5.25	3.186e-15	5.25	5.25

batch	porosity_mode	c_rate	n	cap_mean	cap_std	cap_p5	cap_p95
Batch_1	corrected	1	30	5.25	7.906e-15	5.25	5.25
Batch_1	corrected	2	30	5.114	0.0007793	5.113	5.115
Batch_2	measured	0.1	30	5.25	4.873e-15	5.25	5.25
Batch_2	measured	0.5	30	5.25	8.519e-15	5.25	5.25
Batch_2	measured	1	30	5.25	6.322e-15	5.25	5.25
Batch_2	measured	2	30	3.806	0.1609	3.579	3.979
Batch_2	corrected	0.1	30	5.25	6.543e-15	5.25	5.25
Batch_2	corrected	0.5	30	5.25	6.607e-15	5.25	5.25
Batch_2	corrected	1	30	5.25	5.594e-15	5.25	5.25
Batch_2	corrected	2	30	5.114	0.0006796	5.112	5.114
Batch_3	measured	0.1	30	5.25	7.779e-15	5.25	5.25
Batch_3	measured	0.5	30	5.25	8.37e-15	5.25	5.25
Batch_3	measured	1	30	5.25	6.978e-15	5.25	5.25
Batch_3	measured	2	30	3.92	0.1654	3.641	4.155
Batch_3	corrected	0.1	30	5.25	7.529e-15	5.25	5.25
Batch_3	corrected	0.5	30	5.25	6.698e-15	5.25	5.25
Batch_3	corrected	1	30	5.25	5.9e-15	5.25	5.25
Batch_3	corrected	2	30	5.114	0.0004331	5.113	5.114

